Lecture 10: Sequences

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Understand what a sequence is and how to write one using correct notation.
- ☐ Know what is meant for a sequence to converge or diverge.
- ☐ Find the limit of a sequence using different rules and techniques.
- ☐ Use the Sandwich Theorem and Continuous Function Theorem to evaluate limits.
- ☐ Recognize if a sequence is increasing, decreasing, or bounded and apply the Monotonic Sequence Theorem whenever possible.

Sequences

A sequence is a list of numbers in given order.

- Finite sequence: $\{a_n\}_{n=1}^{4} = \{2, 4, 6, 8\}$.
- Infinite sequence: $\{a_n\}_{n=1}^{\infty} = \{1, 4, 9, 16, \ldots\}$.

We can also see a sequence as a function whose domain is the set of positive integers. In our course, unless specified otherwise, we assume all sequences are real-valued.

Representing Sequences

Instead of writing $a(n)$ for the function value, we write a_n to denote the nth term of the sequence.

Instead of $a : \mathbb{N} \to \mathbb{R}$, we denote a sequence by $\{a_n\}$ or $\{a_n\}_{n=1}^{\infty}$ or $\{a_1, a_2, a_3, \ldots\}$.

Sequences can be described by writing rules that specify their terms (the nth term of the sequence).

Eg:

1. $\{a_n\} = \{2, 4, 6, 8, \ldots\}$
2. $\{a_n\} = \{1, 4, 9, 16, \ldots\}$
3. $\{a_n\} = \{1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \frac{1}{5}, \ldots\}$
4. $\{a_n\} = \{2, 2, 2, 2, \ldots\}$

Convergence and Divergence

The sequence $\{a_n\}$ converges to the number L if for every positive number ϵ there corresponds an integer N such that

$$|a_n - L| < \epsilon \text{ whenever } n > N.$$

If no such number L exists, we say that $\{a_n\}$ diverges.

If $\{a_n\}$ converges to L, we write $\lim_{x\to\infty} a_n = L$ or $a_n \to L$, and we say that L is the limit of the sequence.

Example

Show that $\lim_{n\to\infty} \frac{1}{n} = 0$.

Example

Show that the sequence $\{\frac{n-1}{n+1}\}_{n=1}^{\infty}$ converges to 1.

Example

Show that the sequence $\{1, -1, 1, -1, \ldots, (-1)^{n+1}, \ldots\}$ diverges.

Diverges to (Negative) Infinity

In some cases, a sequence diverges because it get larger and larger. As n increases, a_n would become larger than any fixed number. We describe the behaviour of such sequence by the followings

The sequence $\{a_n\}$ diverges to infinity if for every number M there is an integer N such that for all n larger than N, we have $a_n > M$. If this condition holds, then we write

$$\lim_{n\to\infty} a_n = \infty \text{ or } a_n \to \infty.$$

Similarly, if for every number m there is an integer N such that for all n larger than N, we have $a_n < m$, then we say $\{a_n\}$ diverges to negative infinity and write

$$\lim_{n\to\infty} a_n = -\infty \text{ or } a_n \to -\infty.$$

Calculating Limits of Sequences

Suppose that $\{a_n\}$ and $\{b_n\}$ are convergent sequences with $\lim_{n\to\infty} a_n = A$ and $\lim_{n\to\infty} b_n = B$. Then

- $\lim_{n\to\infty} (a_n + b_n) = A + B$
- $\lim_{n\to\infty} (a_n - b_n) = A - B$
- $\lim_{n\to\infty} (k \cdot b_n) = k \cdot B$ (any number k)
- $\lim_{n\to\infty} (a_n \cdot b_n) = A \cdot B$
- $\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{A}{B}$ if $B \neq 0$

Calculating Limits of Sequences

Examples:

(a) $\lim_{n\to\infty}(-\frac{1}{n})$

(b) $\lim_{n\to\infty}(\frac{n-1}{n})$

(c) $\lim_{n\to\infty}\frac{5}{n^2}$

(d) $\lim_{n\to\infty}\frac{4-7n^6}{n^6+3}$

(e) $\lim_{n\to\infty}(\sqrt{n^2+4n}-n)$

Calculating Limits of Sequences

Remark:

1. One consequence of the previous theorem is that if both $\{a_n\}$ and $\{b_n\}$ converges, their sum $\{a_n + b_n\}$ also converges. However, the converse is not true.

2. Another consequence is that every nonzero multiple of a divergent sequence $\{a_n\}$ diverges.

Calculating Limits of Sequences

The Sandwich Theorem for Sequences

Suppose that $\{a_n\}$, $\{b_n\}$ and $\{c_n\}$ are sequences such that $a_n \leq b_n \leq c_n$ holds for all n beyond some index N. If $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$, then $\lim_{n\to\infty} b_n = L$.

Example:

- $\frac{\cos n}{n} \to 0$
- $\frac{1}{2^n} \to 0$
- $(-1)^n \frac{1}{n} \to 0$
- If $|a_n| \to 0$, then $a_n \to 0$
- $\frac{n!}{n^n} \to 0$

Calculating Limits of Sequences

The Continuous Function Theorem for Sequences

Suppose that $\lim_{n\to\infty} a_n = L$ and that f is continuous at L. Then

$$\lim_{n\to\infty} f(a_n) = f(L)$$

Example: Find $\lim_{n\to\infty} \sin\left(\frac{\pi n^2}{4n^2+1}\right)$, if it exists.

Using L'Hôpital's Rule

Theorem: Suppose that $f(x)$ is a function defined for all $x \geq n_0$ and that $\{a_n\}$ is a sequence such that $a_n = f(n)$ for $n \geq n_0$. Then

$$\lim_{n\to\infty} a_n = L \text{ whenever } \lim_{x\to\infty} f(x) = L.$$

Note: This only works for finite limits.

Example

Show that $\lim_{n\to\infty} \dfrac{\ln n}{n} = 0.$

Example

Suppose that $a_n = (1 + \frac{1}{n})^n$. Determine the limiting behaviour of a_n as $n \to \infty$.

Commonly Occurring Limits

1. $\lim_{n\to\infty} \frac{\ln n}{n} = 0$
2. $\lim_{n\to\infty} \sqrt[n]{n} = 1$
3. $\lim_{n\to\infty} x^{1/n} = 1 \ (x > 0)$
4. $\lim_{n\to\infty} x^n = 0 \ (|x| < 1)$
5. $\lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n = e^x \ (\text{any } x)$
6. $\lim_{n\to\infty} \frac{x^n}{n!} = 0$

Bounded Monotonic Sequences

A sequence $\{a_n\}$ is bounded from above if there exists a number M such that $a_n \leq M$ for all n. Such number M is called an upper bound for $\{a_n\}$.

A sequence $\{a_n\}$ is bounded from below if there exists a number M such that $a_n \geq M$ for all n. Such number M is called a lower bound for $\{a_n\}$.

If $\{a_n\}$ is bounded from above and below, then $\{a_n\}$ is bounded. Otherwise, we say $\{a_n\}$ is an unbounded sequence.

Bounded Monotonic Sequences

A sequence $\{a_n\}$ is nondecreasing if $a_n \leq a_{n+1}$ for all n.

A sequence $\{a_n\}$ is nonincreasing if $a_n \geq a_{n+1}$ for all n.

The sequence $\{a_n\}$ is monotonic if it is either nondecreasing or nonincreasing.

Bounded Monotonic Sequences

The Monotonic Sequence Theorem

If a sequence $\{a_n\}$ is both bounded and monotonic, then the sequence converges.

Bounded Monotonic Sequences

Let the sequence $\{a_n\}$ be defined recursively by

$$a_0 = 1, \quad a_{n+1} = \frac{1}{2}(a_n + 3) \quad \text{for all } n \geq 0.$$

Evaluate $\lim_{n\to\infty} a_n$, giving reasons.